MILITARY SPECIFICATION

CONTAINER RACK, QUADRUPLE (QUADCON RACK)

This specification is approved for use within the United States Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the quadruple container rack (QUADCON rack).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Code LMA-2, Washington, D.C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7125

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SPECIFICATIONS

FEDERAL

QQ-A-200	Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded; General Specification for

MILITARY

MIL-C-5541	Chemical Conversion Coatings on Aluminum Alloys
MIL-Q-9858	Quality Program Requirements
MIL-C-29445(MC)	Container, Quadruple (QUADCON)
MIL-C-29446(MC)	Container, Insert (INSERT)

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

MARINE CORPS

835042A0000	Container Rack, Quadruple (QUADCON Rack)

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order a sample shall be subjected to first article inspection (see 4.4 and 6.3). The first articles shall be manufactured using the same prodedures and tooling proposed for production. Approval of the first articles by the acquiring activity shall not relieve the Contractor of the obligation to supply equipment conforming to this specification. Changes or deviations in production units from the approved first article baseline shall require written authorization by the acquiring activity.

3.2 Materials. Materials and components shall conform to the requirements specified herein and in Drawing No. 835042A0000. Materials not specified shall be of the same quality used for the intended purpose in commercial practice. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly.

3.2.1 Raw materials. The raw materials used in the fabrication of the QUADCON rack shall be in accordance with the applicable drawings. The aluminum rail extrusions and structurals shall be in accordance with QQ-A-200, with the alloys in accordance with the applicable drawings.

3.2.2 Dissimilar materials. Intimate contact of dissimilar materials, as defined in MIL-STD-889, which can be expected to cause galvanic corrosion, shall be avoided. When such contact cannot be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

3.3 Design and construction.

3.3.1 QUADCON Rack. The QUADCON rack shall be fabricated in accordance with Drawing No. 835042A0000. All columns and cross structures and longitudinal support rails shall be fabricated from aluminum structures and extrusions conforming to 3.2.1. No part of the QUADCON rack, including all fasteners, shall protrude from

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the rack to create a human safety hazard or cause chafing of the insert container within the rack. The design of the QUADCON rack shall facilitate ease of installation within the QUADCON. The QUADCON rack assembly shall be prefabricated to the maximum extent possible to reduce the amount of assembly required within the QUADCON.

3.3.2 Supporting rails. The QUADCON rack shall incorporate continuous support rails to provide uniform longitudinal support under the side panels of the insert container. The sliding surfaces of the rails shall be continuous and flat and all securing fasteners shall be recessed to avoid chafing and abrasion of the insert containers.

3.3.3 Container, QUADCON. When the QUADCON rack is fully erected within the QUADCON, built in accordance with MIL-C-29445(MC), 36 insert containers built in accordance with MIL-C-29446(MC) shall fit within the QUADCON with the QUADCON doors closed. Testing shall be in accordance with 4.7.1.

3.3.4 Interchangeability. Like demountable hardware components of the QUADCON rack shall be interchangeable.

3.3.5 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and punching shall be done neatly and accurately. Corners shall be square and true. All bends shall be made with controlled means in order to ensure uniformity of size and shape.

3.3.6 Bolted and riveted connectors. Bolt and rivet holes shall be accurately punched or drilled and have the burrs removed. Washers, lockwashers, or locknuts shall be provided where necessary and all bolts, nuts, and screws shall be tight. Rivet heads, when not countersunk or flattened, shall be of uniform size and shape for the same diameter rivet, concentric with the rivet holes, and in full contact with the surface of the members.

3.3.7 Welding. The surface of the parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Weld penetrations shall be such as to provide transference of maximum design stress through the base metal juncture. Weld configurations and sizes shall be as specified on the applicable drawings.

3.4 Maintainability. The QUADCON rack has been designed so that as much of the maintenance, as possible, may be performed as far forward (preferably by the using unit).

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3.4.1 Organizational level maintenance. Preventive maintenance (PM) consists primarily of inspecting for cleanliness, mechanical integrity and cleaning. PM is performed by designated Marines in the supply section at the smallest unit level. Corrective maintenance, such as replacing minor parts, is performed by the unit's organic maintenance personnel. No specific MOS has been designated for second echelon maintenance; only common tools in maintenance work spaces are required.

3.4.2 Intermediate Level Maintenance. Intermediate level maintenance (3d and 4th echelons) is performed by the Maintenance Battalion, FSSG. Maintenance will include testing the alignment of structural members and realigning as required, replacing major parts, welding and performing modification as directed. Typical MOS could be 1316 (metal worker), 1341 (Engineer equipment mechanic). and 2161 (machinist).

3.4.3 Depot Level Maintenance. No depot level maintenance is required for the QUADCON rack.

3.5 Transportability. The QUADCON rack shall be capable of supporting 36 loaded Inserts inside the QUADCON when the QUADCON is being transported in the Marine, highway, rail, or air mode of dry cargo transportation, military or commercial. The QUADCON rack is not employed separate of the QUADCON. Testing shall be in accordance with 4.7.13, 4.7.15.1 and 4.7.15.2 of MIL-C-29445(MC) and 4.7.6.

3.6 Environmental requirements.

3.6.1 Temperature-humidity. The QUADCON rack shall be capable of functioning and withstanding exposure to temperatures of -65°F to 125°F at any possible relative humidity from 0 to 100 percent. In storage, the QUADCON rack shall be capable of withstanding exposure to temperatures of -70°F to 160°F at any possible relative humidity from 0 to 100 percent without cracking, corroding. delamination, or peeling of protective coatings. Testing shall be in accordance with 4.7.4.1 and 4.7.4.2.

3.6.2 Marine environment. The QUADCON rack shall be fully serviceable when exposed to sea-salt fallout equivalent to 25 pounds/acre/year (2.8 grams/M^2/year). All fasteners, seals, or other hardware shall show no evidence of corrosion or degradation affecting serviceability when tested as required by 4.7.4.2.

3.6.3 Drop shock.

3.6.3.1 Flat drop. The QUADCON rack shall withstand a flat drop of 12 in.(305 mm) from the ground onto a concrete surface. Testing shall be in accordance with 4.7.5.1.

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3.6.3.2 Rotational drop. The QUADCON rack shall withstand rotational drops with one edge placed on a 4 in. (102 mm) block and the opposite edge raised 12 in. (305 mm) and allowed to free fall onto a concrete surface with the 4 in. (102 mm) block acting as a pivot. The QUADCON rack shall withstand this drop on each bottom edge. Testing shall be in accordance with 4.7.5.2.

3.7 Dimensions. The actual external dimensions and their tolerances shall be as listed in Table I. The maximum gross weight rating shall be 5,580 pounds (lbs); however, due to the loading conditions resulting from transportation and handling conditions, the QUADCON rack shall have an ultimate load carrying capacity of 3.75 times 5,580 lbs, or 20,925 lbs. The applicable drawings fully describe all pertinent dimensions and tolerances. Should any dimensions and tolerances result in a component or assembly which will not meet the requirements of this specification, the supplier is responsible for fabrication techniques which will prevent the occurence of such dimensions and tolerances. Testing shall be in accordance with 4.7.2.

TABLE I. Actual External Dimensions and Their Tolerances

Height	Height Tolerances	Width	Width Tolerances	Length	Length Tolerances
(in)	(in)	(in)	(in)	(in)	(in)
69.8	±.06	54.38	±.06	82.25	±.06

3.8 Weight. The tare weight of the QUADCON rack shall be 256 lbs. ± 5 percent. Testing shall be in accordance with 4.7.3.

3.9 Finish. All surfaces shall be prepared as specified on the applicable drawings. All aluminum alloy materials shall be protected by a chemical film conforming to MIL-C-5541 even if a final finish is not applied.

3.10 Identification marking.

3.10.1 Identification. Each QUADCON rack shall have a stenciled part number permanently affixed to the QUADCON rack in accordance with MIL-STD-130.

3.10.2 Markings. Markings shall be in accordance with Drawing Number 835042A0000.

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3.11 Workmanship. All parts, components, and assemblies of the QUADCON rack, including stampings, seals and sealing agents, machined surfaces, and welded parts, shall be clean and free from any defects that will reduce the capability of the QUADCON rack to meet the requirements specified herein. Any components and assemblies which have been repaired or modified to overcome deficiencies or deficient components or assemblies which are determined to be used "as is" shall not be used unless authorized by the contracting activity. External surfaces shall be free from burrs, slag, sharp edges and corners, except where sharp edges and corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the Contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the Contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the Contractor's overall inspection system or quality program in accordance with MIL-Q-9858. The absence of any inspection requirements in this specification shall not relieve the Contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

c. Individual inspection (see 4.6)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.4, 4.5, and 4.6.

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4.4 First article inspection. Prior to beginning production of the QUADCON rack, the Contractor shall fabricate one QUADCON rack and subject it to the tests specified in Table II.

4.5 Quality conformance inspection. From each shipping lot as defined in the contract, a single QUADCON rack shall be chosen at random by the Government and subjected to the tests specified in Table II. Failure of any test shall be cause for inspection of the entire lot at the Contractor's expense for similar defects. Defective QUADCON racks shall not be delivered without correction of the deficiency and then only with the permission of the Contracting Officer.

4.6 Individual inspection. Each QUADCON rack shall be subjected to the inspections and tests specified in Table II. Failure to meet the requirements specified herein shall be cause for rejection. No QUADCON rack shall be reworked and retested without the expressed permission of the contracting activity.

TABLE II. Tests

INSPECTION	REQ PARA	TEST PARA	1ST ART	QUAL CONF	IND
Compatibility	3.3.3	4.7.1	X		
Dimensional	3.7	4.7.2	X	X	X
Weight	3.8	4.7.3	X	X	X
Temperature-Humidity	3.6.1	4.7.4.1 & 4.7.4.2	X		
Marine atmosphere	3.6.2	4.7.4.2	X		
Flat Drop	3.6.3.1	4.7.5.1	X	X	
Rotational Drop	3.6.3.2	4.7.5.2	X	X	
Transportability	3.5	4.7.6	X		

4.7 Tests

4.7.1 Compatibility. A QUADCON built in accordance with MIL-C-29445(MC) and one or more inserts built in accordance with MIL-C-29446(MC) shall be required for this test. Assemble the QUADCON rack into the QUADCON and assure compatibility of 36 insert locations by installing the insert into each location.

4.7.2 Dimensional. The length, width, and height shall be measured and recorded for the QUADCON rack. The requirements of 3.7 shall be satisfied.

4.7.3 Weight. The weight of each QUADCON rack shall be taken. The requirements of 3.8 shall be satisfied.

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4.7.4 Environmental tests.

4.7.4.1 Temperature. The QUADCON rack shall be mounted in a QUADCON with 36 Insert containers emplaced in the QUADCON rack for temperture testing. Low temperature testing shall be in accordance with 4.7.13.3 of MIL-C-29445(MC).

4.7.4.2 Humidity resistance. The QUADCON rack shall be subjected to MIL-STD-810, Method 507.1, Procedure II, excluding steps 2-4, 7, and 8. After cycling, the QUADCON rack shall be examined and show no evidence of delamination, cracking, corrosion, or deterioration of any part of the QUADCON rack.

4.7.4.3 Marine atmosphere. One representative sample of each fastener and track which will be exposed to the atmosphere in the operation and storage modes shall be finished in accordance with the applicable drawings and subjected to tests as described in MIL-STD-810, Method 509.1, except that the salt solution shall have a concentration of 10 percent and the exposure period shall be 96 hours. Upon completion of this test, the test items shall show no evidence of delamination, cracking, corrosion, or deterioration.

4.7.5 Shock tests.

4.7.5.1 Flat Drop. The QUADCON rack shall be assembled into the QUADCON and shall be loaded with 36 inserts to the rated gross weight of 5,580 lbs. (24,820 Newtons). The QUADCON shall be lifted 12 in. (305 mm) above a flat concrete surface. The distance between the concrete surface and each corner of the QUADCON under test shall not vary more than 0.5 in.(12.7 mm) at the time of the drop. The QUADCON shall then be allowed to free fall to the concrete surface. Upon completion of this test, the QUADCON rack shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.2 shall then be satisfied.

4.7.5.2 Rotational drop. The QUADCON rack shall be assembled into the QUADCON and shall be loaded with 36 inserts to the rated gross weight of 5,580 lbs. (24,820 Newtons). The QUADCON shall be tested with one edge supported 4 in. (102 mm) above a flat concrete surface. The opposite edge shall be lifted a distance of 12 in. (305 mm) above the concrete surface and allowed to free fall onto the surface. This procedure shall be repeated until each edge has been allowed to free fall. Upon completion of this test, the QUADCON rack shall show neither permanent deformation nor abnormality with will render it unsuitable for use and the dimensional requirements of 4.7.2 shall then be satisfied.

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4.7.6 Transportability. The QUADCON rack shall be mounted in the QUADCON with 36 Insert containers emplaced in the QUADCON rack for testing in accordance with 3.5.3, 4.17.15.1 and 4.17.15.2 of MIL-C-29445(MC).

4.8 Test reports. The results of all testing shall be reported and written in compliance with MIL-STD-831 fully describing and illustrating the test procedures and test results.

5. PACKAGING

5.1 Preservation. The QUADCON rack requires no additional preservation.

5.2 Packing. Unless otherwise specified in the contract, no protective media need be installed prior to shipment. Conventional methods of blocking, bracing, and strapping are to be employed during shipment by rail, road, ship, and air as described in MIL-STD-1186.

5.3 Marking. In addition to the marking required by 3.10, each QUADCON rack shall be marked for shipment in accordance with the requirements of MIL-STD-129. A removable data plate shall be installed to show cube, weight, and destination specified by the contracting activity.

6. NOTES

6.1 Intended use. The QUADCON rack covered by this specification is intended to provide support for 36 insert containers in a six-high by three-wide by two-long arrangement for use as bins/drawers when the QUADCON is used in its optional bin/drawer configuration.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article testing is required (see 3.1).

c. When a quality control system is required (see 4.1.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List

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(CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Para No.	Data Requirement Title	Applicable DID No.	Option

(Data item descriptions related to this specification, and identifed in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. First Article test items shall be 36 sample items manufactured using the same procedures and tooling proposed for production. The Contractor shall submit a First Article test plan and receive Government approval prior to testing. The First Articles shall be examined and tested for approval at the Contractor's plant or at an independent laboratory acceptable to the Government. First Article tests shall be witnessed by a Government representative of the acquiring activity. After completion of First Article testing, the Contractor shall submit a separate test report. The Government reserves the right to waive the requirement for First Article testing for items that have previously been tested and accepted by the Government.

6.4 Subject term (key word) listing. The following subject term (key word) listing is furnished for retrieval searches.

Container Rack

Container Rack, Quadruple

QUADCON Rack

Preparing activity
Navy-MC
Project No. 7125-N120

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